

INVESTIGATIONS OF SYNTACTIC AMBIGUITY*¹

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SUMMARY

Two experiments were conducted with male and female university students in an attempt to establish whether syntactically ambiguous words are better represented by independent lexical entries for each syntactic context in which the items occur, or whether the different usages could be adequately represented by single entries. Both experiments were attempts to influence recall through control of the coding of ambiguous items. Experiment one ($N = 128$) involved a free recall procedure in which ambiguous items were embedded in lists of nonambiguous nouns or verbs, and Experiment two ($N = 144$) involved a release from proactive inhibition procedure which has been shown to make syntactic features highly salient. The results of both experiments supported the single entry view; that is, the ambiguous items apparently were not coded according to one usage independent of other usages.

A. INTRODUCTION

A number of theorists have been engaged in the development of computer models which are intended to describe the representation of words in the internal lexicon (e.g., 2, 3). One important consideration for any such model should be the representation of words with more than one usage. That is, should an ambiguous word, such as *bank*, have only a single entry, or should there be a separate entry for each usage?

In the case of semantic ambiguity (i.e., words with more than one distinct meaning), it seems likely that multiple entries for ambiguous words are required. Rubenstein, Lewis, and Rubenstein (6), for example, found that *Ss* could identify semantically ambiguous words as English words significantly faster than they could identify nonambiguous words, which suggests that the

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decision for an ambiguous word simply requires locating any one usage during a search of the lexicon. With regard to syntactically ambiguous words (i.e., words that occur in more than one syntactic context), however, Rubenstein *et al.* found no differences in identification time between ambiguous and nonambiguous words. Thus, Rubenstein *et al.* concluded that semantically ambiguous items have more entries in the lexicon than syntactically ambiguous or nonambiguous items, and that the latter two do not differ in their number of entries.

The Rubenstein *et al.* data for syntactically ambiguous words seem somewhat surprising, since many theorists have assumed a different representational scheme for different syntactic classes of words. Therefore, the present study was an attempt to investigate syntactic ambiguity further with the use of different experimental procedures than those employed by Rubenstein *et al.* The specific question of concern was whether a syntactically ambiguous word is better represented with an independent lexical entry for each syntactic context in which the word occurs, or whether the different contextual occurrences can be represented in a single entry. (For present purposes, the latter alternative is taken to include the possibility of multiple entries, all of which are activated if any one entry is activated.)

In order to test the possibility of independent lexical entries, two experiments were conducted involving attempts to force Ss to encode ambiguous words according to only one usage. A free recall paradigm was employed in Experiment 1, in which lists were constructed such that either 15 nouns or 15 verbs were combined either with five items of the other class (i.e., nouns with verbs or vice versa) or with five syntactically ambiguous words. Experiment 2 involved a variation on the release from proactive inhibition (PI) paradigm, in which *E*-provided cues were employed to bias the encoding of ambiguous items.

B. EXPERIMENT 1

Prior work (5) demonstrated that nouns are recalled better than verbs in a free recall paradigm, and it was predicted that such would be the case in the present experiment. It was also expected that the encoding of ambiguous items would be determined by syntactic context and that, if ambiguous items have independent entries, there would be differential recall of those items in different contexts. Thus, it was expected that the probability of recalling the word *judge* (which can occur as either a noun or a verb) would be influenced by the overall syntactic structure of the list in which it occurred.

1. *Method*

Two practice lists of 20 items each were constructed involving proper nouns and adjectives. After these lists were presented for free recall, *Ss* received one of four test lists, involving either 15 nouns and five verbs (NV), 15 nouns and five ambiguous items (NA), 15 verbs and five nouns (VN), or 15 verbs and five ambiguous items (VA). From the groups of 15 nouns and 15 verbs, five of each were selected to allow statistical comparisons in the four conditions. The remaining 10 items of each type served as the first five and the last five items of the lists, in order to eliminate primacy and recency effects and to establish a dominant syntactic context. The order of the primacy and recency items in the list was random, and the only restriction on the 10 critical items was that no more than two items from a particular set (e.g., nouns or verbs, nouns or ambiguous words, etc.) occurred consecutively. Two different presentation orders were obtained for both the critical and the noncritical items to minimize any ordering effects.

To control for item effects, the total number of syllables in each set was constant (a total of eight syllables for each comparison set of five items and 18 for the sets of 15 noncritical items), and the mean frequencies of the items in the various sets was 57.00 for ambiguous items, 60.20 for critical nouns, 59.60 for critical verbs, 73.30 for noncritical nouns, and 70.60 for noncritical verbs (4). Standard instructions for free recall were employed, and the lists were presented orally by a tape recorder at a 2.5 second rate. Each list began with the word "ready," and the end of the list was indicated by a brief sound from a buzzer. Two minutes were allowed for recall.

Male and female students in introductory psychology at Florida State University served as *Ss* in partial fulfillment of a course requirement. There were 32 *Ss* in each of the four conditions—run in groups of six or less—making a total of 128 *Ss*.

2. *Results and Discussion*

The mean number of correct responses for critical items is reported in Table 1. It is apparent that there was no substantial influence due to dominant context. Nouns were recalled better than verbs in all conditions where that comparison was made, and ambiguous items were recalled better than either nouns or verbs.

In order to perform a factorial analysis, dominant context items (i.e., critical items of the same form as the initial and final items) were compared to nondominant context items (i.e., critical items not of the same form as the

TABLE 1
MEAN NUMBER OF CORRECT RESPONSES IN EXPERIMENT 1

Syntactic class	Condition			
	NV	NA	VN	VA
Nouns	1.34	1.25	1.59	—
Verbs	.97	—	1.03	1.06
Ambiguous	—	1.97	—	2.19

Note: NV = 15 nouns and five verbs; NA = 15 nouns and five ambiguous items; VN = 15 verbs and five nouns; VA = 15 verbs and five ambiguous items.

initial and final items) as a within-Ss factor, and experimental condition was taken as a between-Ss factor. Analysis of variance indicated that the main effect of experimental condition was not significant [$F(3, 124) = 2.59, p > .10$], while the dominant-nondominant context factor was reliable [$F(1, 124) = 17.01, p < .01$]. It is impossible to draw any particular conclusions from these main effects, however, and the source of variance of greatest interest is the interaction of experimental condition with dominant-nondominant context [$F(3, 124) = 6.64, p < .01$].

In order to examine the interaction, simple effects were calculated for the pair of means in each experimental condition. This analysis revealed reliable differences in the NA ($p < .01$), VA ($p < .01$), and VN ($p < .05$) conditions, with recall of ambiguous items greater than recall of nouns or verbs and recall of nouns greater than recall of verbs. Although the difference in recall between nouns and verbs in condition NV was not reliable ($p > .05$), the difference was in the same direction as that for condition VN. Thus, while the recall of nouns was significantly higher than that of verbs in at least one condition, the most striking aspect of these data is the markedly higher level of recall for ambiguous items relative to either nouns or verbs.

To further compare the recall of the three types of items, a Newman-Keuls analysis was calculated for the four sets of nondominant context items (nouns or ambiguous items in a verb context and verbs or ambiguous items in a noun context). This analysis also indicated that nouns and ambiguous items were recalled better than verbs, but despite the fact that the mean recall of ambiguous items was noticeably higher than that of nouns, these differences failed to achieve significance. Of primary importance, however, is the lack of a significant difference between the recall of ambiguous items in a noun context and the recall of the same items in a verb context. This result is clearly inconsistent with the notion of independent lexical entries for syntactically ambiguous items.

The unexpected recall superiority of ambiguous items over that of nouns requires some explanation. One possibility is that there are multiple lexical entries for ambiguous items, but that the entries are not independent and both entries are tagged at input, increasing the probability of a successful search at recall. Another possible explanation is that ambiguous items are entered only once in the lexicon, but since ambiguous words have generally been employed in more varied contexts than have nonambiguous words, there are more coding possibilities for ambiguous words. Thus, in a sense, an entry for an ambiguous item would contain more information than entries of nonambiguous items, which could facilitate processing and retrieval of the former. Both of these possibilities seem reasonable, but the single entry view is perhaps more consistent with the data reported by Rubenstein *et al.*

It must be noted that standard frequency tables, such as Kučera and Francis (4) do not distinguish among the various usages of ambiguous words. Since frequency was approximately equal for the various conditions in the present study, it must be that the frequency of any one usage of an ambiguous item is less than the combined frequency. To the extent that high frequencies lead to increased recall, the results of Experiment 1 would be a conservative estimate of the advantage ambiguous items have over nonambiguous nouns and verbs, since the latter two classes have higher mean frequencies than any one usage of the ambiguous items. Of course, if the single entry view is correct, this problem may be irrelevant, since the different occurrences of ambiguous items are contained within a single entry.

Experiment 2 was simply an attempt to create a different and possibly stronger test of the independent lexical entry hypothesis. In addition, it tested this hypothesis under conditions in which different levels of recall for different classes of items was not a factor.

C. EXPERIMENT 2

In a recent study, Bird and Goodwin (1) demonstrated effective use of syntactic information in a release from proactive inhibition (PI) paradigm. This paradigm basically involves several trials on a Brown-Peterson task, in which word triads organized along some dimension (e.g., some semantic grouping) are presented as test material. Recall of word triads typically decreases over trials (presumably due to PI), but may increase dramatically on, say, Trial 4 with a change along the dimension of organization (e.g., to a different semantic grouping—see Wickens (7) for a review of this work).

Although some studies have found syntactic organization to be ineffective in producing release from PI (7, 8), Bird and Goodwin found substantial

release when syntactic information was made highly salient. That is, when *Ss* were informed before each trial as to whether nouns or verbs would be involved in that trial, a change in organization from nouns to verbs or vice versa on Trial 4 produced reliable release from PI.

This procedure was employed in the present study to examine further the possibility of independent lexical entries for ambiguous words. Three trials involving either noun triads or verb triads were followed by a triad of syntactically ambiguous words. Before each of the first three trials the *E* announced the syntactic organization of the triad by saying "nouns" or "verbs." On Trial 4, release or no-release conditions were created, depending on whether the *E* called the ambiguous items nouns or verbs. If there are independent lexical entries for the two usages of the ambiguous items the *E*'s announcement should bias coding toward one entry, leading to release from PI if the items are biased toward the class not included on the first three trials (i.e., if the *E* calls the ambiguous items verbs after three trials of nouns, recall should increase substantially on Trial 4).

1. Method

There were three major experimental conditions: release, in which the ambiguous items were biased toward the syntactic class not involved in the first three trials; no-release, in which the bias was toward the same class as that involved in the first three trials; and a nonsalient control, in which no information was provided about the organization of the triads. The experimental conditions were combined factorially with the syntactic class of the first three triads (nouns or verbs) yielding six different cells. Finally, there were also six possible orders of presentation for the first three triads and two different triads of ambiguous items which could be presented on Trial 4, making a total of 72 cells in the design.

The entire matrix was replicated twice with one male and one female *S* in each cell. Thus, there was a total of 144 *Ss* (24 in each of the six conditions), assigned to conditions randomly with the restriction that all six conditions were run with a given order of presentation before the order was changed. Students at Florida State University served as *Ss* in partial fulfillment of a course requirement.

The words selected to serve as nonambiguous experimental items were single function nouns and verbs (i.e., none of the nouns ever occur as verbs and vice versa—e.g., *rice* or *shut*), while the ambiguous items selected were judged by the *E* to be easily employed in sentences as either nouns or verbs (e.g., *drill* or *brush*). There was no obvious semantic similarity among the

words and no overlap of initial phonemes within the triads seen by a given *S*. All items were one syllable words and were assigned to triads randomly with the restriction that the mean frequencies of the triads were comparable (ranging from 27.3 to 39.3).

The triads were printed in block letters on 2×2 slides with the use of a vertical array with each item shifted one space to the right of the item above [see Wickens (7)]. The interval between the presentation of a triad and recall was filled with an arithmetic task involving single digits (0-9), selected randomly and printed in block letters on 2×2 slides in the form of a 5×5 array.

Instructions were read by the *S* and were of two types depending on the experimental condition. The *Ss* in the release and no-release conditions were told that the triads consisted of either nouns or verbs and that the *E* would indicate before each trial whether the upcoming triad would involve nouns or verbs. However, no suggestion was made to the *S* that he should attempt to use this information to assist him in remembering the triads. For *Ss* in the nonsalient condition no information was provided concerning the organization of the triads. The slides were presented by a Kodak Carousel projector on a screen about three feet in front of the *S*. All *Ss* were tested individually, and presentation of the slides was controlled by the *E*, with the aid of a stopwatch.

Each trial began with a two-second presentation of a slide which showed three asterisks. This served as a cue for the presentation of the triad and provided an opportunity to inform *Ss* in the release and no-release conditions whether the triad would consist of nouns or verbs (the *E* simply said the word "nouns" or "verbs"). The triad was then presented for 1.5 seconds, and the *S* read each word aloud once. Following the triad, a number array was presented for 20 seconds during which time the *S* said each number and its product when multiplied by three, working as quickly as possible from left to right across each row. Finally, a slide with three question marks was presented for 8.5 seconds during which time the *S* tried to recall the triad aloud. This procedure continued for a total of four trials.

2. Results and Discussion

Responses were scored as correct when they were given within the recall interval of the trial on which they first occurred. Table 2 presents the percentage of correct responses as a function of trials for the three main conditions, each entry representing the combined recall of nouns and verbs. Recall generally decreased over trials, and there is no indication of any release from PI on Trial 4 for the release condition. A $3 \times 2 \times 4$ analysis of variance was calculated in which the factors were experimental condition, syntactic class of

TABLE 2
PERCENTAGE OF CORRECT RESPONSES IN EXPERIMENT 2

Condition	Trials			
	1	2	3	4
Release	92	51	53	43
No-release	90	61	42	42
Nonsalient	87	57	51	44

the first three triads, and trials. The only significant factor in the entire analysis was the trials factor [$F(3, 414) = 63.21, p < .001$]. Thus there was no reliable interaction of condition with trials [$F(6, 414) = 1.12, p > .20$], indicating no reliable release from PI. The syntactic class factor was also not reliable [$F(1, 138) < 1$], nor did syntactic class interact with any other factor.

The recall of nouns and verbs was comparable in all conditions, as indicated by the absence of a three-way interaction [$F(6, 414) = 1.12, p > .20$]. This finding suggests that the ambiguous items were truly ambiguous, since, for example, if they were predominantly nouns, release should have been obtained when the first three triads were verbs, at least in the release condition. There were no trends in the data to suggest this may have been the case.

D. GENERAL DISCUSSION

The data of Experiment 2 are clearly inconsistent with an independent entry view of syntactically ambiguous words. The combined data of the two experiments reported here, as well as that from Rubenstein *et al.*, make a reasonably strong case in favor of a single entry view of syntactically ambiguous words, although the high level of recall for ambiguous items in Experiment 1 is consistent with a structure in which there are multiple lexical entries, with all entries being activated when any one entry is activated.

If it is accepted that syntactically ambiguous words have only a single entry in the lexicon, then it is necessary to determine whether a general model of language structure should employ a similar base structure for all syntactic classes, or whether ambiguous items are some sort of special case. If the latter position is taken, then it must be determined whether ambiguous items have a "core" entry of noun, verb, adjective, etc., with a tag which permits usage in other contexts, or whether the entry is truly ambiguous, with the necessary features to permit multiple usage more or less equally present. In any case, it seems possible that a better understanding of ambiguous words could aid in the development of models of language structure.

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